

Stress-Testing Object Detection on MAN TruckScenes: A Condition-Stratified Evaluation

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Abstract—MAN TruckScenes is the first large-scale multimodal dataset for autonomous trucking, yet its 34 scene-level condition tags remain unexploited for systematic robustness analysis. We use these tags to perform a condition-stratified evaluation of YOLOv8 object detection across weather, lighting and area-type conditions using all four cameras. We project the dataset's 3D bounding boxes into 2D YOLO format, train a baseline detector and disaggregate performance by scene condition to identify where detection degrades most notably. Our results show that a clear-weather-only detector fails to generalise to adverse conditions, with rain producing the largest degradation. Class rebalancing through oversampling improves rare-class detection for some classes but does not transfer to condition-stratified robustness or better overall performance compared to the Baseline model.

Keywords—object detection, autonomous trucking, YOLO, stress-testing, condition analysis, adverse weather, MAN TruckScenes

1. INTRODUCTION

Perception systems for autonomous driving have advanced noticeably in recent years, driven by large-scale datasets and deep learning architectures. However, this progress has been mainly focused on passenger vehicles. The trucking domain introduces perception challenges, such as elevated and wider sensor mounting positions, trailer occlusions and operation in terminal environments with dense, slow-moving traffic, that are distinct from passenger vehicles [4]. While significant progress has been made in autonomous driving perception, these truck-specific factors remain largely unexplored due to the lack of dedicated datasets.

The release of MAN TruckScenes at NeurIPS 2024 marked the first large-scale multimodal dataset specifically designed for autonomous trucks. Each scene is annotated with condition tags describing weather, lighting, road structure and area type.

The dataset authors, Fent et al. [4], provide an initial per-condition analysis of 3D detection baselines, reporting reduced performance under rain, fog and tunnel conditions. We extend this analysis to 2D detection with YOLOv8, examine the underlying condition and class distributions that drive these gaps and evaluate across all four camera positions. Our goal is not to propose a new architecture but to provide a detailed empirical picture of where a standard detector breaks down on this dataset and what that implies for real-world trucking perception.¹

2. RELATED WORK

2.1. Autonomous Driving Datasets

Autonomous driving research has advanced considerably in the past decade, originating with KITTI in 2012 as the first multimodal benchmark for the domain. Liu et al. [5] present a comprehensive survey of 265 such datasets, evaluating them across sensor modalities, annotation quality and contextual conditions. This survey, published in IEEE Transactions on Intelligent Vehicles in 2024, does not include MAN TruckScenes, which appeared at NeurIPS 2024 shortly after, and none of the catalogued datasets target autonomous trucking specifically.

TruckScenes [4] extends the multimodal approach established by nuScenes [1] to the heavy truck domain for the first time, including the first 4D radar data with full 360° coverage. Its devkit follows the nuScenes format, making it easier to adopt for existing research.

2.2. Robustness of Object Detection Under Adverse Conditions

Object detection performance degrades under adverse weather and lighting, that is a well-documented challenge for autonomous driving systems. Mirza et al. [2] perform one of the most detailed evaluations of single- and dual-modality detection architectures on data captured in real weather conditions, demonstrating significant performance drops in rain, fog and snow compared to clear scenes. Their work highlights the importance of evaluating detectors under the specific conditions that are to be encountered in deployment, rather than relying entirely on aggregate metrics.

On the solution side, Chu [3] proposes D-YOLO, a dual-route framework that integrates dehazing directly into the YOLO detection pipeline at the feature level, showing improved performance on hazy and foggy benchmarks. The existence of such specialised architectures accentuates the severity of the problem. Standard detectors struggle enough under adverse conditions that dedicated architectures are developed to address it.

3. RESEARCH QUESTIONS

Our analysis is structured around three questions that the three camera-based model training runs and the per-condition / per-camera evaluation jointly address.

RQ1: Where does YOLOv8 detection degrade?

Across which weather conditions and camera positions does YOLOv8 lose the most performance on TruckScenes? How much of the performance gap is driven by structural challenges (class composition, camera placement) and how much by visual difficulty (different weather conditions)?

RQ2: Does clear-weather generalise to adverse conditions?

Can a detector trained exclusively on clear-weather scenes generalise to rain, fog, snow, and overcast?

RQ3: Does class-balance transfer to condition robustness?

Does tiered oversampling of underrepresented classes improve per-class AP? Does the same intervention affect condition-stratified gaps in any way?

4. DATASET

4.1. MAN TruckScenes Overview

MAN TruckScenes [4] was collected using a MAN TGX 18.510 long-haul truck equipped with four cameras, six LiDAR sensors, six 4D radar sensors, two IMUs and a high-precision RTK-GNSS unit. The dataset comprises 747 scenes of approximately 20 seconds each, recorded across highways, logistics terminals, rural roads and urban areas in Germany. Annotations consist of manually labelled 3D bounding boxes at 2 Hz, covering 27 object classes mapped to 12 detection classes using the official devkit mapping. In this work we focus on the camera modality, which Fent et al. [4] identified as the most condition-sensitive sensor in their baseline evaluation (0.02 mAP vs 0.27 for LiDAR). The four cameras produce images at 1980 × 943 pixels, yielding 95,608 sample-camera pairs across the trainval split.

¹Full implementation code available at <https://github.com/emacstud/deep-learning-group-project>

4.1.1. Scene Condition Distribution

Each scene carries tags across seven condition categories. Tables 1 and 2 report scene and image counts for the train and validation splits.

Table 1. Scene and image counts — weather, area, and daytime conditions. Image counts shown in parentheses.

Category	Condition	Train	Val	Total
Weather	Clear	225 (35,972)	39 (6,240)	264 (42,212)
	Overcast	155 (24,800)	23 (3,660)	178 (28,460)
	Rain	109 (17,436)	10 (1,600)	119 (19,036)
	Snow	22 (3,520)	1 (160)	23 (3,680)
	Fog	11 (1,740)	1 (160)	12 (1,900)
	Other	1 (160)	1 (160)	2 (320)
Area	Highway	392 (62,672)	49 (7,840)	441 (70,512)
	Terminal	46 (7,360)	8 (1,280)	54 (8,640)
	Rural	44 (7,040)	8 (1,260)	52 (8,300)
	City	27 (4,316)	8 (1,280)	35 (5,596)
	Residential	11 (1,760)	1 (160)	12 (1,920)
	Parking	3 (480)	0 (0)	3 (480)
Daytime	Noon	281 (44,956)	38 (6,060)	319 (51,016)
	Morning	203 (32,432)	25 (4,000)	228 (36,432)
	Evening	39 (6,240)	12 (1,920)	51 (8,160)

Table 2. Scene and image counts — season, lighting, and structure conditions. Image counts shown in parentheses.

Category	Condition	Train	Val	Total
Season	Summer	250 (39,972)	22 (3,520)	272 (43,492)
	Autumn	191 (30,536)	34 (5,440)	225 (35,976)
	Winter	82 (13,120)	19 (3,020)	101 (16,140)
Lighting	Illuminated	431 (68,912)	49 (7,820)	480 (76,732)
	Twilight	33 (5,280)	11 (1,760)	44 (7,040)
	Dark	29 (4,640)	9 (1,440)	38 (6,080)
	Glare	29 (4,636)	5 (800)	34 (5,436)
	Other	1 (160)	1 (160)	2 (320)
Structure	Regular	335 (53,548)	50 (7,980)	385 (61,528)
	Underpass	97 (15,520)	12 (1,920)	109 (17,440)
	Overpass	55 (8,800)	9 (1,440)	64 (10,240)
	Bridge	27 (4,320)	3 (480)	30 (4,800)
	Tunnel	9 (1,440)	1 (160)	10 (1,600)

The distribution is heavily skewed toward dominant conditions. Clear weather accounts for 44.2% of all images, highway driving for 73.8%, and illuminated lighting for 80.2%. Rare conditions such as fog (2.0%), tunnel (1.7%), and snow (3.9%) are substantially underrepresented. The official validation split contains only one scene each for fog, snow, and tunnel, making it unreliable for condition-stratified assessment. This motivated the construction of a dedicated stratified holdout set.

4.1.2. Overlapping Conditions

Overlapping conditions reveals several structural dependencies that constrain evaluation. Table 3 shows the joint distribution of weather and daytime conditions.

Table 3. Scene counts by weather and daytime (train + val combined).

Weather	Evening	Morning	Noon
Clear	47	111	106
Overcast	0	76	102
Rain	4	28	87
Snow	0	0	23
Fog	0	12	0

Three overlapping conditions are particularly relevant. Firstly, every fog scene in the dataset is also tagged as morning, so the fog and morning labels co-occur perfectly within our data and we cannot separate the contribution of fog from that of morning daytime. Secondly, all 34 glare scenes also carry the rain tag, so the glare and rain la-

bels overlap completely and our evaluation cannot distinguish their effects. Lastly, dark scenes are predominantly recorded under clear weather (33 of 38 scenes), with none under fog or snow, which limits any per-weather analysis of the dark lighting condition.

4.1.3. Annotation Density by Condition

Table 4 reports the mean number of annotated boxes per image by condition. This matters because annotation density correlates strongly with mAP independent of visual degradation — a scene with fewer objects will tend to yield higher mAP regardless of model quality.

Table 4. Mean annotated boxes per image by condition.

Category	Condition	Mean boxes / image
Weather	Overcast	13.3
	Rain	13.1
	Clear	10.3
	Fog	8.7
	Snow	7.9
Daytime	Evening	13.1
	Morning	11.6
	Noon	11.4
Area	City	23.5
	Parking	14.0
	Residential	13.0
	Highway	11.6
	Rural	9.2
	Terminal	6.4

The density variation across conditions is substantial and directly relevant to interpreting evaluation results. City scenes average 23.5 boxes per image compared to 6.4 in terminal and 7.9 in snow scenes. Notably, clear weather — the most represented condition at 44.2% of images — has a below-average density of 10.3 boxes per image, lower than both rain (13.1) and overcast (13.3). This means raw mAP comparisons across conditions reflect differences in scene geometry and object density as much as weather-driven visual degradation, and must be interpreted accordingly.

4.1.4. Class Composition by Condition

Beyond aggregate density, class composition varies substantially across areas. Figure 1 shows that terminal scenes are dominated by trucks (30.9%) and trailers (30.2%), while city and highway scenes are car-dominated (51%). Pedestrians are concentrated in city (7.3%) and terminal (9.1%) scenes and nearly absent from highway (0.2%). This means per-area mAP differences would reflect not only detection difficulty but also which classes are present to be detected.



Figure 1. Class Composition by Area Type

4.1.5. Camera Position Analysis

The four cameras are not equivalent in annotation density. Table 5 reports the mean number of annotated boxes per image by camera position.

Table 5. Mean annotated boxes per image by camera position.

Camera	Mean boxes / image
Left Front	14.9
Right Front	14.6
Left Back	11.6
Right Back	5.4

The right back camera consistently has the lowest annotation density across all weather conditions. This reflects two truck-specific factors, i.e. trailer occlusion, since the right back camera faces the trailer body, and right-hand traffic in Germany, where right-side cameras primarily face the road shoulder rather than adjacent traffic lanes. The left back camera, facing adjacent-lane and oncoming traffic, captures roughly twice as many objects as the right back camera. The front cameras (CAMERA_LEFT_FRONT at 14.9 and CAMERA_RIGHT_FRONT at 14.6) capture the highest number of objects overall. This density asymmetry is structural rather than condition-dependent.

5. METHODS

5.1. 3D to 2D Projection

The MAN TruckScenes dataset provides annotations as 3D bounding boxes defined in the world coordinate frame, that follows the nuScenes format. Since YOLOv8 operates on 2D images, a projection pipeline was needed to convert these 3D annotations into 2D bounding boxes which would be suitable for training. This section describes the coordinate transformation chain and the filtering decisions applied.

5.1.1. Coordinate Transformation Chain

Each 3D bounding box is defined by its own centre point (x, y, z) , dimensions (w, l, h) , and orientation that are expressed as a quaternion (q_w, q_x, q_y, q_z) . Internally, `get_sample_data()` applies a chain of rigid-body transformations to map each annotation from the world frame to the ego-vehicle frame and then to the camera's local frame, returning the resulting 3D corner coordinates together with the camera intrinsic matrix K . Using the devkit method rather than re-implementing the transform chain avoids the risk of frame-convention errors at any link in the chain.

Given the eight corners of a 3D box expressed in the camera frame, projection onto the image plane follows the standard pinhole model:

$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} \sim K \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} \quad (1)$$

where (u, v) are the resulting pixel coordinates. The 2D bounding box is then taken as the axis-aligned bounding rectangle of all the different projected corner points.

5.1.2. Handling Boxes That Straddle the Camera Plane

A non-trivial case arises when a 3D box partially crosses the camera plane, which means that some corners have $Z > 0$ while others have $Z \leq 0$. Simply discarding corners behind the camera produces incorrect bounding boxes for objects that are partially in front of and partially behind the image plane, which occurs at close range or at the image periphery.

To handle this, each of the twelve edges of the box is checked. Where an edge crosses the near-plane threshold $Z = 0.1$, a new point is interpolated along that edge at the crossing position:

$$\mathbf{p}_{\text{clip}} = \mathbf{p}_a + t(\mathbf{p}_b - \mathbf{p}_a), \quad t = \frac{Z_{\min} - Z_a}{Z_b - Z_a} \quad (2)$$

where $Z_{\min} = 0.1$ is the near-plane depth threshold. These clipped intersection points are added to the point set before projection, ensuring the resulting 2D box correctly covers the visible portion of the object.

5.1.3. Filtering Criteria

Three filtering conditions were applied to remove the labels that would be uninformative or harmful to training.

First, any box whose projected corners all fall behind the camera ($Z \leq 0.1$ for all eight corners) is entirely discarded, as it has no visible presence in the image.

Second, projected boxes are removed whose post-clipping width or height is below 4 pixels. At the native camera resolution of 1980×943 pixels, such boxes correspond to objects at distances beyond the reliable annotation range and would contribute only noise to the training signal.

Third, for the boxes that do not straddle the camera plane, a visibility fraction check is applied. If the clipped, image-bounded box area is less than 5% of the raw projected area, the box is discarded. This removes objects that are almost entirely outside the image frame, where only a small amount is technically within bounds.

5.1.4. Output Format

The final output of the projection pipeline is a normalised bounding box in YOLO format (c_x, c_y, w, h) , where all values are divided by the image width and height respectively, placing them in the range $[0, 1]$. The pipeline was validated visually by overlaying projected boxes onto raw camera frames across all four camera positions prior to dataset export (Figure 2), confirming pixel-level alignment between the 3D annotations and their 2D projections.

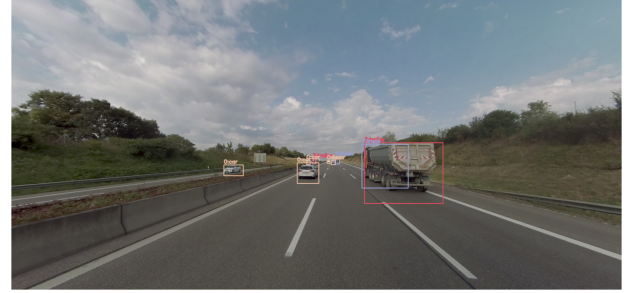


Figure 2. Projection pipeline results showing aligned 3D-to-2D bounding box annotations across Right back camera view.

5.2. Class Mapping

Independently of the projection step, each annotation is mapped from the TruckScenes 27-class taxonomy to the 12 official detection classes via the devkit's `category_to_detection_name()` function. Classes outside the benchmark set – including `vehicle.emergency.*`, `human.pedestrian.stroller`, `human.pedestrian.wheelchair`, and `static_object.bicycle_rack` – are discarded, in line with the official TruckScenes evaluation.

5.3. Training Configuration

All experiments were conducted on Google Colab using a high-RAM runtime with an NVIDIA A100-SXM4-80GB GPU (CUDA 13.0). The base detector used across all three training runs was YOLOv8s, the small variant of the YOLOv8 architecture.

5.3.1. Hardware and Software

- **GPU:** NVIDIA A100-SXM4-80GB (81,920 MiB VRAM)
- **Framework:** Ultralytics YOLOv8s
- **Input resolution:** 960 × 960 pixels
- **Mixed precision:** Automatic Mixed Precision (AMP) enabled

5.3.2. Data Splits

The official MAN TruckScenes train/val split was modified to produce a more reliable evaluation protocol, as the official validation set contains only 1 scene each for fog, snow and tunnel conditions. The official test split is available for download but contains no ground-truth annotations, so it is suitable only for visual inference. Taking all points into consideration, a stratified holdout set was carved from the training data prior to all experiments:

- **Train-keep:** 444 scenes ($\approx 71,000$ images) — official train minus holdout scenes, used for model training
- **Official val:** 75 scenes (11,980 images) — used exclusively for early stopping during training, not for final evaluation
- **Holdout:** 79 scenes ($\approx 12,600$ images) — stratified subset carved from official train, reserved for all final reported metrics; rare conditions (fog, snow, tunnel, dark, twilight, residential) are explicitly represented

5.3.3. Training Runs

Three models were trained, each addressing a distinct research question. Common hyperparameters shared across all runs are listed in Table 6.

Table 6. Shared training hyperparameters across all three runs.

Parameter	Value
Optimiser	AdamW (auto)
LR schedule	Cosine annealing
Mosaic augmentation	1.0
Mixup	0.0
HSV hue / sat / val	0.015 / 0.5 / 0.4
Horizontal flip	0.5
Vertical flip	0.0 (disabled)
Workers	8

Run 1 — Baseline Trained on the full train-keep split across all weather conditions.

- **Training images:** $\approx 71,000$ (444 scenes, all conditions)
- **Batch size:** 32
- **Epochs:** 50
- **Early stopping patience:** 10
- **Purpose:** Reference model trained on the full condition-diverse training set. Supplies the per-condition and per-camera measurements used to characterise where YOLOv8 detection degrades on TruckScenes (RQ1), and serves as the comparison point against which the Clear-Weather (RQ2) and Oversampled (RQ3) runs are evaluated.

Run 2 — Clear-Weather Only Trained exclusively on scenes tagged `tag_weather = clear`.

- **Training images:** $\approx 29,732$ (186 scenes)
- **Batch size:** 32
- **Epochs:** 50
- **Early stopping patience:** 10
- **Purpose:** Tests whether a model trained exclusively on clear-weather scenes generalises to rain, fog, snow, and overcast (RQ2).

Run 3 — Oversampled (Class-Imbalance Correction) Trained on the full train-keep split with tiered oversampling applied to address severe class imbalance. Oversampling factors were computed using inverse frequency relative to the truck class as anchor (143,346 boxes), capped at 6 $\times$ to limit overfitting risk.

- **Training images:** $\approx 71,000$ base images, expanded to $\approx 200,000+$ effective entries per epoch after oversampling
- **Batch size:** 32
- **Epochs:** 50
- **Early stopping patience:** 10
- **Oversampling factors applied:**
 - Trailer: 2 $\times$
 - Traffic cone: 5 $\times$
 - Pedestrian, bus, barrier, motorcycle, bicycle, animal, other vehicle: 6 $\times$ (capped)
- **Purpose:** Tests whether boosting the effective representation of rare classes during training improves their per-class AP and whether class-balanced training transfers to condition robustness (RQ3).

6. RESULTS

6.1. Overall Detection Performance

Table 7 reports the aggregate detection metrics for the trained models evaluated on the stratified holdout set (12,616 images, 79 scenes). The Baseline model achieves the highest overall performance (mAP@0.5: 0.291, mAP@0.5:0.95: 0.142), followed closely by the Oversampled model (0.281 / 0.140). The Clear-Weather Only model scores lowest across both mAP metrics (0.232 / 0.112).

Table 7. Overall holdout validation performance for the trained models.

Split	Epochs	mAP@0.5	mAP@0.5:0.95
Baseline	37	0.291	0.142
Clear-Weather	31	0.232	0.112
Oversampled	16	0.281	0.140

Table 8 reveals a sharp long-tail effect across all three models: the four large-vehicle classes (car, truck, trailer, other vehicle) cluster around $AP@0.5 \approx 0.50$ on the baseline, while the minority classes (bus, pedestrian, motorcycle, animal) remain below 0.12, mirroring the imbalance flagged by the original TruckScenes paper [4]. The clear-weather model degrades on every class relative to the baseline, with two near-collapses – bus drops from 0.115 to 0.019 and animal from 0.097 to 0.001 – indicating that these categories appear almost exclusively in non-clear scenes within the training pool. The oversampled model produces an uneven recovery: motorcycle improves by +38% (0.114 $\rightarrow$ 0.157) and pedestrian and bicycle show small but consistent gains, but this comes at a 2–5 point cost on the common vehicle classes, and neither animal nor barrier recovers to baseline.

Table 8. Per-class AP@0.5 for the trained models.

Class	AP@0.5		
	Baseline	Clear-Weather	Oversampled
Car	0.521	0.474	0.503
Truck	0.523	0.462	0.510
Bus	0.115	0.019	0.109
Trailer	0.535	0.486	0.510
Other vehicle	0.500	0.428	0.501
Pedestrian	0.097	0.056	0.102
Motorcycle	0.114	0.066	0.157
Bicycle	0.268	0.223	0.283
Traffic cone	0.163	0.090	0.142
Barrier	0.268	0.244	0.222
Animal	0.097	0.001	0.064
Traffic sign	0.290	0.235	0.273
Mean (mAP@0.5)	0.291	0.232	0.281

6.2. Condition-Stratified Analysis

The original TruckScenes baselines identify adverse weather as one of the dominant failure modes of camera-based detection. To diagnose where our trained models stand on this axis, we evaluate each model on the holdout split stratified by the `tag_weather` attribute. Table 9 presents the per-weather results across the three trained models.

Per-condition mAP is sensitive to class composition. Each weather subset contains only the classes that appear in scenes annotated with that condition, and the standard mAP averages per-class AP over only those present classes. Fog scenes in our holdout, for example, contain only four classes (car, truck, trailer, traffic sign), while clear scenes evaluate against eleven. The fog number is therefore an average over four relatively easier large-vehicle and sign classes, while the clear number is pulled downward by harder classes (pedestrian, motorcycle, bus, traffic cone, etc.) that simply do not appear in fog. This produces a counter-intuitive artefact in the Raw mAP column: for the Baseline, fog (0.506) appears to outperform clear (0.378), suggesting the model handles fog better than its native condition.

To remove this bias we compute a shared mAP by restricting the per-class average to the set of classes present in every condition within the tag. For weather this common set is {car, truck, trailer, traffic_sign}. The Shared mAP columns of Table 9 normalise every condition's score to this four-class subset, making the values directly comparable across conditions. Under this restriction the artefact inverts: clear (0.541) now outperforms fog (0.506) by 6%. We use Shared mAP as the primary metric for the analyses that follow. The $\Delta\%$ columns report the underperformance of each condition relative to **clear** within the same model.

Under Shared mAP the three models agree at the extremes – clear is easiest and rain is hardest – but diverge in the middle of the gradient. The Baseline and Oversampled models order the conditions **clear** > **fog** > **overcast** > **snow** > **rain**, while the Clear-Weather model orders them **clear** > **overcast** > **fog** > **snow** > **rain**, swapping the relative positions of fog and overcast. Two factors plausibly account for the swap. First, the fog-overcast gap is small in both the Baseline and the Oversampled columns, and fog is the smallest sub-population, so its rank is statistically fragile. Second, overcast is visually closer to clear conditions than fog: it differs from clear primarily in lighting while preserving scene geometry and texture, whereas fog affects long-range objects and blurs edges throughout the scene. A model that has not seen adverse data during training should therefore transfer more cleanly to overcast than to fog, and that is what we observe in the Clear-Weather column. Consistent with this interpretation, the stricter mAP@0.5:0.95 metric ranks overcast above fog for all three models, suggesting that the mAP@0.5 fog rank in the Baseline and Oversampled columns lies at the edge of measurement noise and the **clear** > **overcast** > **fog** > **snow** > **rain** ordering is the more robust description of the underlying difficulty gradient.

Beyond this swap, the broader pattern is unambiguous. The Baseline and Oversampled models drop from 0.541 and 0.523 on clear to 0.305 and 0.292 on rain respectively – both a 44% relative drop – while the Clear-Weather model drops from 0.525 to 0.214, a 59% drop. Notably, rain emerges as a harder condition than snow. This likely reflects the specific visual artefacts of rain (droplets on the windshield, low-contrast wet pavement, optical glare on light sources at night) interfering with feature extraction more than the comparatively uniform brightening of snow. The stricter mAP@0.5:0.95 metric shows the same ordering but with sharper degradation (Baseline and Oversampled both 51% relative drop on rain, Clear-Weather 67%), indicating that adverse conditions also undermine box-localisation precision, not only detection.

6.2.1. Baseline vs. Clear-Weather Model

The Clear-Weather model trails the Baseline in every condition, but the gap is conspicuously asymmetric. On clear weather itself the two

models are within 3% of each other in Shared mAP (Baseline 0.541, Clear-Weather 0.525). As conditions become more adverse the gap widens monotonically: roughly 13% on overcast, 20% on fog, and 30% on both snow and rain. This is the cleanest quantitative evidence in our results: a model trained exclusively on clear-weather scenes does not generalise to adverse conditions, even though it has effectively learnt the clear-weather task itself.

A symmetric observation runs in the opposite direction. The Baseline's 0.541 on clear improves on the Clear-Weather model's 0.525 on clear weather itself, despite the Clear-Weather model having been trained on the data drawn precisely from the clear scenes only. The extra training signal in the Baseline comes entirely from adverse-condition images. The implication is that adverse-condition data acts as a form of implicit regularisation — extending the effective training pool with image-quality variation analogous in effect to colour-jitter, contrast, or blur augmentations — and improves robustness even on the nominally easy condition. Diverse training data is not just helpful on the conditions it covers, but it universally enhanced the performance.

6.2.2. Baseline vs. Oversampled Model

The Oversampled model tracks the Baseline closely across all five weather conditions. Per Shared mAP@0.5, the two models differ by no more than 0.02 absolute on any bucket, with the Baseline consistently scoring marginally higher: clear (0.541 vs 0.523), overcast (0.494 vs 0.474), snow (0.440 vs 0.423), fog (0.506 vs 0.495), and rain (0.305 vs 0.292). The $\Delta\%$ columns confirm that this is a uniform shift rather than a redistribution: the Oversampled model's relative gap from clear matches the Baseline's to within one percentage point on every condition. The same flat pattern holds under the stricter mAP@0.5:0.95 metric.

Our results indicate that class-rebalanced training does not translate into adverse-condition robustness: oversampling neither closes the rain gap nor boosts any specific adverse condition. The Oversampled model trades a small amount of weather-uniform performance for the per-class gains documented in Table 8, where motorcycle, pedestrian, and bicycle AP improve by 5–38% over the Baseline. The implication is that adverse-condition degradation is not bottlenecked by class imbalance. It is driven by the underlying distribution of visual artefacts that class-balancing cannot address. Recovering rain performance specifically would require either condition-balanced training data or architectural interventions targeted at adverse-condition robustness.

6.3. Camera Comparison

Table 10 reports per-camera mAP on the holdout. Each camera contributes an equal 3,154 images. The Shared columns restrict the per-class average to the eleven detection classes that appear across all four cameras (animal is absent from Left Back).

A clear structural pattern emerges across all three models. Left Back is the highest-performing camera, with the other three positions trailing by 17–35%. Right Front records the largest gap in every model, while Right Back and Left Front cluster in the middle: Right Back leads Left Front for the Baseline and the Clear-Weather model, but the order swaps marginally for the Oversampled model. Two factors plausibly account for this. First, back cameras observe predominantly close-range traffic, while front cameras face long-range traffic at substantially higher object density – the per-camera annotation analysis (Section 4.1.5) reports roughly twice-thrice as many ground-truth boxes per image in the front cameras. Small, far-field objects are intrinsically harder to detect, which penalises both front cameras. Second, Right Back is doubly disadvantaged. The truck's trailer occupies part of its field of view, reducing the effective frame area. In addition, in a right-hand-traffic country such as Germany – where the dataset was collected – the right side of the vehicle predominantly faces the road shoulder rather than adjacent lanes or oncoming traffic.

Table 9. Per-condition holdout performance across the three trained models (Baseline, Clear-Weather, Oversampled). The *Shared* columns restrict the per-class average to the classes present in *every* condition value within the weather tag, removing class-composition bias from raw mAP. The $\Delta\%$ columns report the relative gap to **clear** (lower is closer to clear) condition within the same tag and model.

Condition	Model	n	Classes	mAP@0.5			mAP@0.5:0.95		
				Raw	Shared	$\Delta\%$	Raw	Shared	$\Delta\%$
fog	Baseline	620	4	0.506	0.506	6%	0.272	0.272	11%
	Clear-Weather			0.403	0.403	23%	0.196	0.196	33%
	Oversampled			0.495	0.495	5%	0.263	0.263	11%
clear	Baseline	6,240	11	0.378	0.541	—	0.193	0.305	—
	Clear-Weather			0.360	0.525	—	0.183	0.291	—
	Oversampled			0.387	0.523	—	0.204	0.294	—
snow	Baseline	1,120	9	0.292	0.440	19%	0.143	0.226	26%
	Clear-Weather			0.172	0.306	42%	0.080	0.145	50%
	Oversampled			0.283	0.423	19%	0.138	0.215	27%
overcast	Baseline	2,240	12	0.235	0.494	9%	0.117	0.276	10%
	Clear-Weather			0.216	0.428	18%	0.099	0.224	23%
	Oversampled			0.239	0.474	9%	0.118	0.264	10%
rain	Baseline	2,396	9	0.185	0.305	44%	0.090	0.150	51%
	Clear-Weather			0.122	0.214	59%	0.055	0.097	67%
	Oversampled			0.167	0.292	44%	0.081	0.144	51%

Left Back is also subject to partial trailer occlusion and consequently lower per-image annotation density, but it benefits from a view of the adjacent lanes and oncoming traffic, partially offsetting the reduced frame area.

The asymmetry is robust to model choice. The four cameras’ relative ranking is identical in both Baseline and Clear-Weather, and the within-camera variation between the three models (11–22%) is uniformly smaller than the within-model variation across cameras (approximately 34%). Per-camera differences are therefore structural — driven by camera geometry, trailer occlusion, and the typical distribution of object distances each camera observes. Any architectural or training-strategy intervention aimed at adverse-condition robustness will inherit this same ordering, and per-camera mAP should be interpreted with the asymmetry in mind: a higher mAP at Left Back does not necessarily indicate stronger detection at that camera in absolute terms, but rather an easier intrinsic detection task.

6.4. Loss Functions

Figure 3 shows the training and validation loss curves for all three runs across box, classification (cls) and distribution focal (dfl) loss components.

Baseline (37 epochs, Figure 3a). All three loss components descend smoothly. Validation loss plateaus by approximately epoch 10–15 and stabilises while training loss continues to fall, producing a moderate but consistent generalisation gap. The stable plateau with no divergence indicates the model is learning a well-regularised solution on

the full condition-diverse training set.

Clear-Weather only (31 epochs, Figure 3b). A sharp spike appears in all three validation losses during epochs 2–4, absent in the baseline. This early instability is consistent with reduced data diversity: training exclusively on clear scenes narrows the gradient landscape, causing unstable updates when the model encounters the stylistic variety of the official validation split. The validation plateau is reached at a higher loss value than the baseline across all three components, reflecting the larger generalisation gap already quantified in Table 7.

Oversampled (16 epochs, Figure 3c). Early stopping fires after only 16 of the 50 scheduled epochs. Training loss descends steeply (oversampling inflates the effective dataset size, producing a fast-moving training signal), but validation loss, evaluated on the un-oversampled official validation split, plateaus within the first 3–4 epochs. This distribution mismatch between the oversampled training pool and the standard validation set causes the patience threshold to trigger well before the model has finished learning, likely underestimating the true benefit of oversampling.

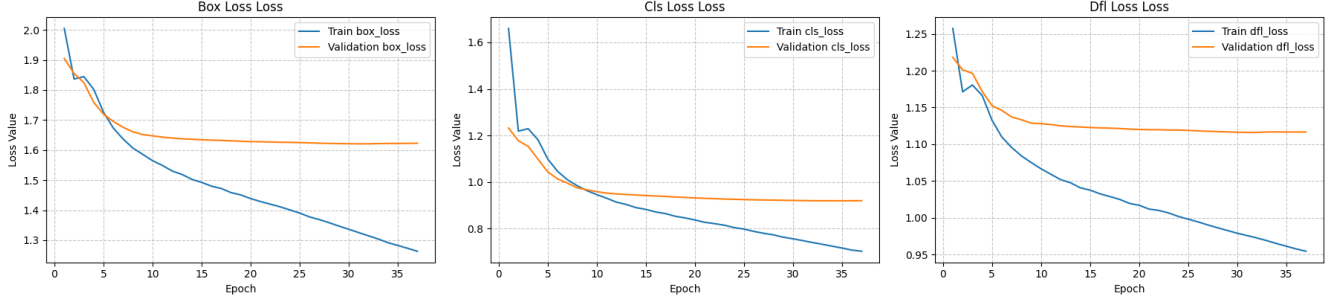
7. DISCUSSION

Our condition-stratified evaluation produced three empirical findings. First, the largest performance gap on the weather axis is the rain condition: all three models drop 44–59% relative to clear under Shared mAP@0.5. Second, a Clear-Weather-only training pipeline fails to generalise to adverse conditions, but the converse also holds – the Baseline outperforms the Clear-Weather model even on clear scenes, suggest-

Table 10. Per-camera holdout performance across the three trained models. The *Shared* columns restrict the per-class average to the 11 classes common across all four cameras (every detection class except animal, which has no ground-truth instances in Left Back). The $\Delta\%$ columns report the underperformance relative to the within-model reference camera (Left Back, the highest-performing position).

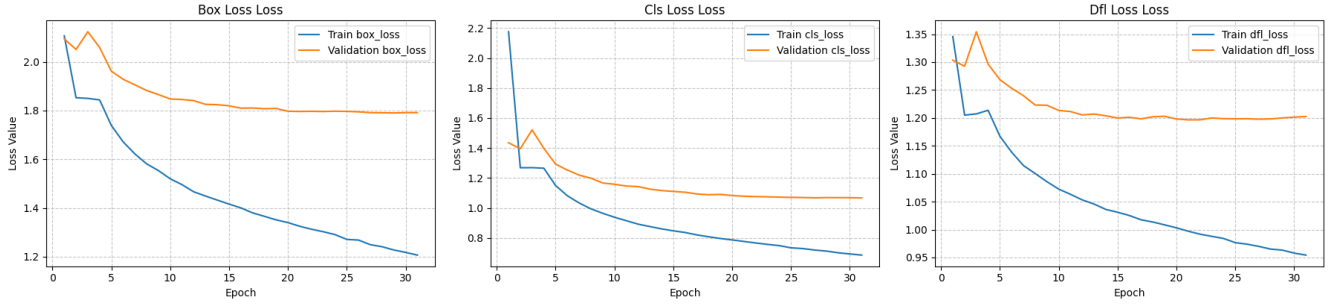
Model	Camera	n	Classes	mAP@0.5			mAP@0.5:0.95		
				Raw	Shared	$\Delta\%$	Raw	Shared	$\Delta\%$
Baseline	Left Back	3,154	11	0.403	0.403	—	0.207	0.207	—
	Left Front	3,154	12	0.264	0.286	29%	0.130	0.154	32%
	Right Back	3,154	12	0.289	0.307	24%	0.143	0.141	26%
	Right Front	3,154	12	0.262	0.266	34%	0.129	0.130	37%
Clear-Weather	Left Back	3,154	11	0.332	0.332	—	0.172	0.172	—
	Left Front	3,154	12	0.206	0.224	33%	0.097	0.105	39%
	Right Back	3,154	12	0.251	0.274	17%	0.117	0.128	26%
	Right Front	3,154	12	0.199	0.217	35%	0.094	0.102	41%
Oversampled	Left Back	3,154	11	0.389	0.389	—	0.208	0.208	—
	Left Front	3,154	12	0.262	0.284	27%	0.128	0.139	33%
	Right Back	3,154	12	0.262	0.277	29%	0.130	0.140	33%
	Right Front	3,154	12	0.245	0.258	34%	0.122	0.126	39%

Baseline Results - Training vs. Validation Losses



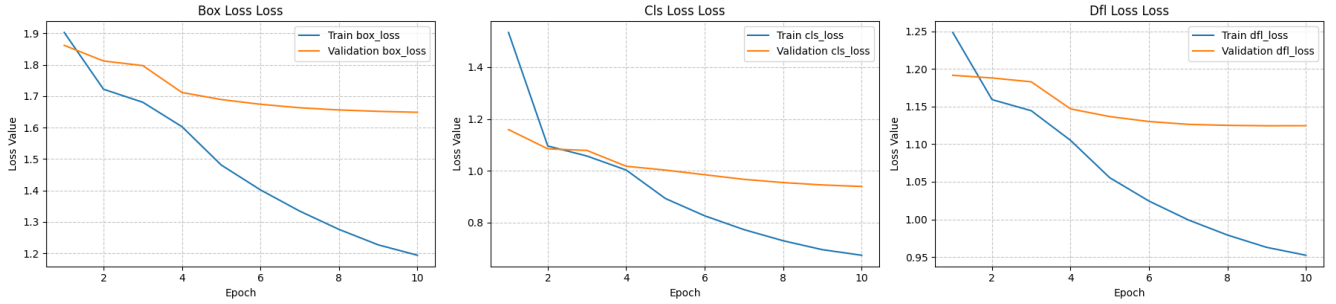
(a) Baseline (37 epochs): smooth convergence, moderate train-val gap.

Clear Results - Training vs. Validation Losses



(b) Clear-Weather only (31 epochs): early spike in epochs 2–4, higher validation plateau.

Oversample Results - Training vs. Validation Losses



(c) Oversampled (10 epochs): steep training descent, early stopping triggered by distribution mismatch between oversampled training data and standard validation split.

Figure 3. Training vs. validation loss curves (box, cls, dfl) for the three trained models. Orange: validation loss. Blue: training loss.

ing that adverse-condition images act as implicit regularisation. Third, class-rebalancing through oversampling produces per-class gains (notably +38% on motorcycle) but does not affect condition-stratified performance or increase the overall performance above Baseline model, indicating that adverse-condition degradation and class imbalance are independent problems.

7.1. Performance Gaps and Their Sources

The largest gaps emerge along two different angles, and our targeted training-side intervention failed to close either. On weather (Table 9), rain produces the steepest degradation. Rain-specific visual artefacts – windshield droplets, low-contrast wet pavement, glare – interfere with feature extraction more aggressively than the comparatively uniform effect of snow, despite both being adverse conditions.

On camera position (Table 10), Right Front records a 34% Shared mAP gap to Left Back across all three models – a structural asymmetry that the architecture and training strategy cannot completely mitigate, driven by the camera’s view of the outer side of the road rather than active lanes in right-hand-traffic Germany, and by trailer occlusion.

Inverse-frequency oversampling, our intended remedy for the long

tail of rare classes, recovered some class-level performance (Table 8) but produced essentially no change in condition-stratified mAP: the Oversampled model tracks the Baseline within 0.02 absolute Shared mAP@0.5 on every weather bucket.

Each observation points to a different type of failure mode: rain is a feature-extraction problem, the Right Front asymmetry is a sensor-placement problem, and class-balanced training addresses the class-imbalance problem without affecting the condition-imbalance problem.

7.2. Limitations and Future Work

Several factors constrain the generalisability of our findings. We evaluated a single detector architecture (YOLOv8s), while architectural choices such as RT-DETR or transformer-based detectors may exhibit different condition-sensitivity patterns. We used only keyframe annotations (2 Hz), exploiting no temporal information: video-based or recurrent feature-aggregation models might recover information lost in single-frame detection under adverse conditions, where individual frames are degraded but multi-frame aggregation is informative. Condition tags in TruckScenes are scene-level rather than per-frame, and

several condition pairs are intrinsically correlated in the dataset (fog appears only in morning scenes, glare primarily with rain, dark scenes predominantly with clear weather), which prevents disentangling the independent contribution of each condition without additional diverse data.

These limitations point to several concrete directions for future work. The most immediate is 4D radar fusion: radar is the dataset's primary novel contribution per Fent et al. [4], and integrating radar features into a fusion architecture (CRN or BEVFusion variants) would directly test whether the condition-dependent camera degradation we documented is recoverable through complementary sensor modalities. LiDAR, increasingly common in object-detection pipelines, would extend this direction further – camera, radar, and LiDAR streams are all available in TruckScenes. A condition-balanced training experiment – the direct analogue of our Oversampled run, targeting condition imbalance rather than class imbalance – would clarify whether condition-balanced training closes the rain gap. Temporal models that aggregate features across keyframes within a scene would test whether single-frame detection limits are recoverable through context. Finally, given that clear-weather training does not generalise to adverse conditions, a complementary experiment would test the converse direction: whether training restricted to adverse-condition scenes generalises back to clear weather.

7.3. Implications

Three practical implications follow for autonomous trucking perception systems. First, training-data diversity matters more than quantity for adverse-condition robustness. The Clear-Weather model – trained exclusively on clear weather images drawn from the complete distribution – underperformed the Baseline on clear weather itself, which argues against curating training sets to remove difficult scenes. The difficult scenes are exactly the regularisation signal the model needs. Second, class-balanced and condition-balanced training address independent problems. Practitioners targeting adverse-condition performance need explicit condition-balanced sampling or domain-adaptation techniques, not just class rebalancing – the 2–4% uniform shift our Oversampled model exhibits across all weather buckets demonstrates that rare-class augmentation provides no adverse-condition robustness on its own. Third, per-camera confidence calibration is warranted: a 34% mAP gap between Right Front and Left Back across all three models suggests that detection thresholds should not be uniform across cameras, and downstream tracking systems consuming multi-camera detections should weight them by position-specific reliability.

8. CONCLUSIONS

We conducted a condition-stratified evaluation of YOLOv8 object detection on MAN TruckScenes, training three detectors that differ only in training-data composition – a baseline trained on all conditions, a clear-weather-only variant, and a class-rebalanced (oversampled) variant – and evaluating each on a stratified holdout disaggregated by weather and camera position. Three empirical findings emerged. Clear-weather training fails to generalise to adverse conditions, with the largest degradation on rain. Adverse-condition images function as implicit regularisation: the Baseline outperforms the Clear-Weather model even on clear weather, indicating that diverse training data is universally beneficial rather than only on the conditions it covers. And class-rebalancing through oversampling improves rare-class detection without positively affecting condition-stratified or overall performance, demonstrating that class imbalance and condition imbalance are independent problems requiring distinct interventions.

For autonomous trucking perception, the combination of rain-condition degradation and the structural gap between front and back cameras argues against relying on camera-only detection in safety-critical adverse-weather operation. Closing these gaps will require interventions tailored to each – sensor fusion or domain-adaptive

training for the weather gap, per-camera confidence calibration for the sensor-placement asymmetry – rather than the architectural or training-time fixes that effectively address class imbalance alone.

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